

Initiating Search

November 10, 2025, 2:48 PM

• Search:

CAS Lexicon Search:

Sharpless dihydroxylation - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (338)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (487)	 Reactions	View Results
Filtered By:		
Yield:	90-100%, 80-89%, 70-79%	
Document Type:	Journal	

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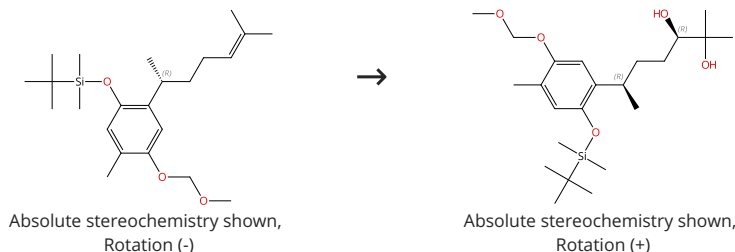
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Reactions (50)

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Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%



31-277-CAS-7281582

Steps: 1 Yield: 100%

Highly efficient, enantiocontrolled total syntheses of (+)-heliannuol D and (-)-helibisabonol A

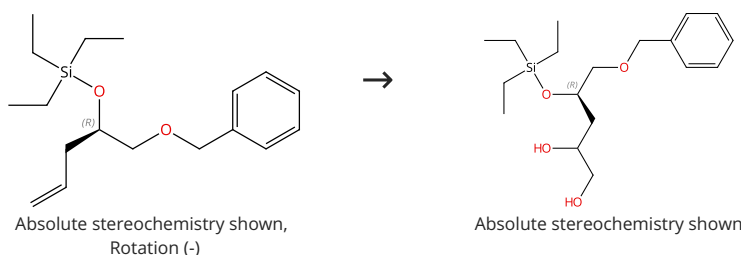
By: Manabe, Yuki; et al

Heterocycles (2014), 88(1), 441-452.

1.1 **Reagents:** Methanesulfonamide, 1,4-Bis(9-*O*-dihydroquinidiny)phthalazine**Solvents:** *tert*-Butanol, Water; 0 °C; 15 h, rt1.2 **Reagents:** Sodium sulfite; 0 °C; 1 h, rt

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%



31-614-CAS-23944326

Steps: 1 Yield: 100%

Asymmetric Total Synthesis of Amphirionin-2

By: Saha, Dhiman; et al

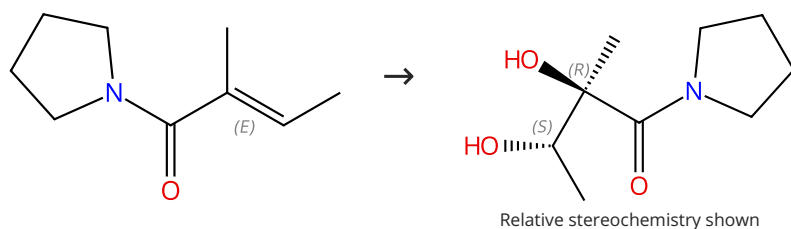
Journal of Organic Chemistry (2021), 86(15), 10006-10022.

1.1 **Reagents:** N-methylmorpholine N-oxide**Catalysts:** Osmium tetroxide**Solvents:** *tert*-Butanol, Tetrahydrofuran, Water; 0 °C; 12 h, rt; rt → 0 °C1.2 **Reagents:** Sodium bisulfite**Solvents:** Water

Experimental Protocols

Scheme 3 (1 Reaction)

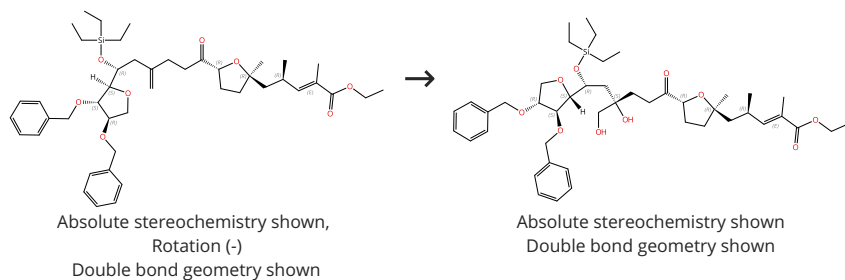
Steps: 1 Yield: 100%


 Suppliers (3)

31-277-CAS-1584789 Steps: 1 Yield: 100%	Asymmetric Dihydroxylation of Esters and Amides of Methacrylic, Tiglic, and Angelic Acid: No Exception to the Sharpless Mnemonic!
1.1 Reagents: Citric acid, N-methylmorpholine N-oxide Catalysts: Potassium osmate dihydrate Solvents: <i>tert</i> -Butanol, Water; rt	By: Weber, Fabian; et al
1.2 Reagents: Sodium sulfite Solvents: Water; rt	European Journal of Organic Chemistry (2015), 2015(11), 2428-2449.
Experimental Protocols	

Scheme 4 (1 Reaction)

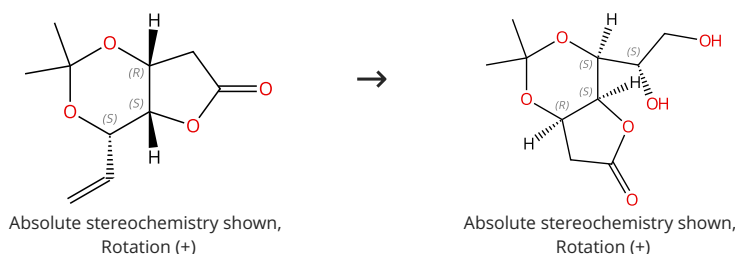
Steps: 1 Yield: 100%



31-277-CAS-20967853 Steps: 1 Yield: 100%	Rhodium-catalyzed vinyl 1,4-conjugate addition coupled with Sharpless asymmetric dihydroxylation in the synthesis of the CDE ring fragment of pectenotoxin-4
1.1 Reagents: Potassium carbonate, Methanesulfonamide, Potassium osmate dihydrate Catalysts: Potassium ferricyanide, 1,4-Bis(9- <i>O</i> -dihydroquinidiny)phthalazine Solvents: <i>tert</i> -Butanol, Water; 0 °C; 19 h, 0 °C → rt	By: Richardson, Melodie S. W.; et al Chemical Science (2019), 10(25), 6336-6340.
1.2 Reagents: Sodium sulfite Solvents: Water; rt	

Scheme 5 (1 Reaction)

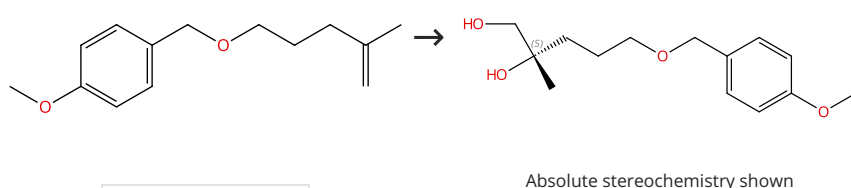
Steps: 1 Yield: 100%



31-277-CAS-19794224 Steps: 1 Yield: 100%	Asymmetric synthesis of the cytotoxic lactone (+)-cardiobutanolide and two novel analogues
1.1 Reagents: N-methylmorpholine N-oxide Catalysts: Osmium tetroxide Solvents: Acetone, <i>tert</i> -Butanol, Water; 23.5 h, rt	By: Kovacevic, Ivana; et al Tetrahedron Letters (2019), 60(9), 684-687.
1.2 Reagents: Sodium sulfite; 1 h, rt Experimental Protocols	

Scheme 6 (1 Reaction)

Steps: 1 Yield: 100%

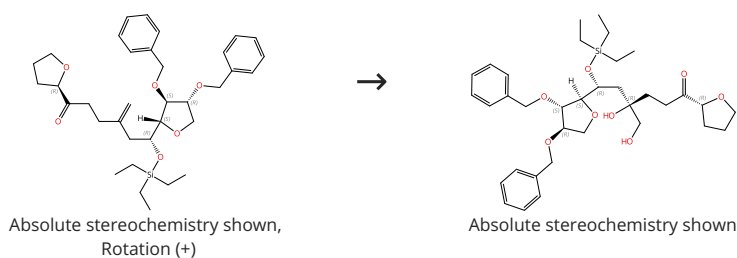


Suppliers (4)

31-277-CAS-7716859 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate, Methanesulfonamide, Potassium ferricyanide Catalysts: Potassium osmate dihydrate, 1,4-Bis[(9<i>S</i>)-10,11-dihydro-6'-methoxycinchonan-9-yl]-9,10-anthracenedione Solvents: <i>tert</i>-Butanol, Water; 2 h, 0 °C 1.2 Reagents: Sodium thiosulfate; 0 °C; 0 °C → rt; 60 min, rt Experimental Protocols	Asymmetric Sharpless dihydroxylation reaction of chiral bishomoallylic alcohols: Application to the synthesis of the C1-C10-C5 fragment of FR225654 By: Mohammad, Shabbair; et al Synlett (2013), 24(19), 2581-2585.
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Scheme 7 (1 Reaction)

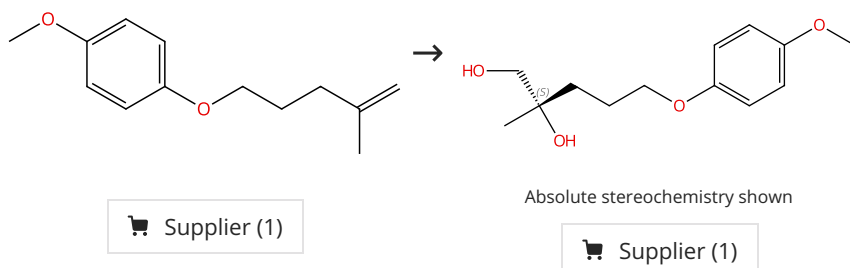
Steps: 1 Yield: 100%



31-277-CAS-20967841 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate, Methanesulfonamide, Potassium osmate dihydrate Catalysts: Potassium ferricyanide, (DHQ)₂PHAL Solvents: <i>tert</i>-Butanol, Water; 0 °C; 19 h, 0 °C → rt 1.2 Reagents: Sodium sulfite Solvents: Water; rt	Rhodium-catalyzed vinyl 1,4-conjugate addition coupled with Sharpless asymmetric dihydroxylation in the synthesis of the CDE ring fragment of pectenotoxin-4 By: Richardson, Melodie S. W.; et al Chemical Science (2019), 10(25), 6336-6340.
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Scheme 8 (1 Reaction)

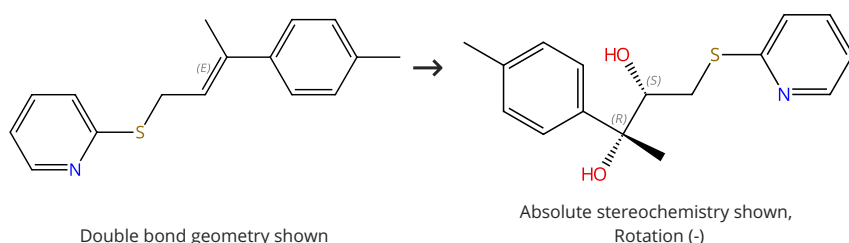
Steps: 1 Yield: 100%



31-277-CAS-11974339 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate, Methanesulfonamide, Potassium ferricyanide Catalysts: Potassium osmate dihydrate, 1,4-Bis[(9<i>S</i>)-10,11-dihydro-6'-methoxycinchonan-9-yl]-9,10-anthracenedione Solvents: <i>tert</i>-Butanol, Water; 2 h, 0 °C 1.2 Reagents: Sodium thiosulfate; 0 °C; 0 °C → rt; 60 min, rt Experimental Protocols	Asymmetric Sharpless dihydroxylation reaction of chiral bishomoallylic alcohols: Application to the synthesis of the C1-C10-C5 fragment of FR225654 By: Mohammad, Shabbair; et al Synlett (2013), 24(19), 2581-2585.
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Scheme 9 (1 Reaction)

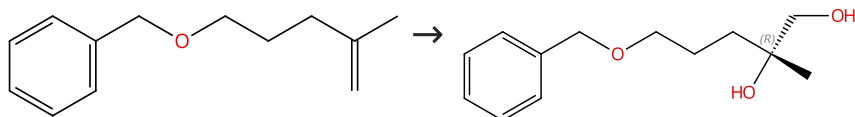
Steps: 1 Yield: 100%



31-277-CAS-24350122	Steps: 1 Yield: 100%	Practical synthesis of aromatic bisabolanes: Synthesis of 1,3,5-bisabolatrien-7-ol, peniciaculin A and B, and hydroxysydonic acid
1.1 Reagents: Methanesulfonamide, AD-mix-β Solvents: <i>tert</i> -Butanol, Water; 0 °C; overnight, 4 °C		By: Yajima, Arata; et al
1.2 Reagents: Sodium sulfite Solvents: Water		Tetrahedron (2021), 92, 132253.
Experimental Protocols		

Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown

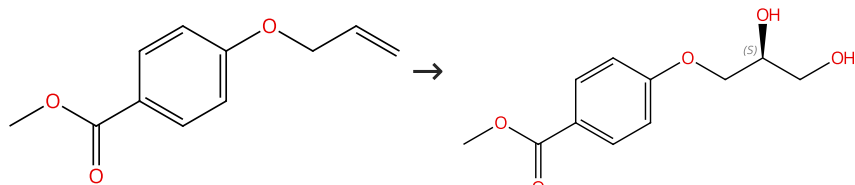
Suppliers (3)

Suppliers (4)

31-277-CAS-5587092	Steps: 1 Yield: 100%	Asymmetric Sharpless dihydroxylation reaction of chiral bishomoallylic alcohols: Application to the synthesis of the C1-C10-C5 fragment of FR225654
1.1 Reagents: Potassium carbonate, Methanesulfonamide, Potassium ferricyanide Catalysts: Potassium osmate dihydrate, (DHQ) ₂ PHAL Solvents: <i>tert</i> -Butanol, Water; 4 h, 0 °C		By: Mohammad, Shabbair; et al
1.2 Reagents: Sodium thiosulfate; 0 °C; 0 °C → rt; 60 min, rt		Synlett (2013), 24(19), 2581-2585.
Experimental Protocols		

Scheme 11 (1 Reaction)

Steps: 1 Yield: 99%

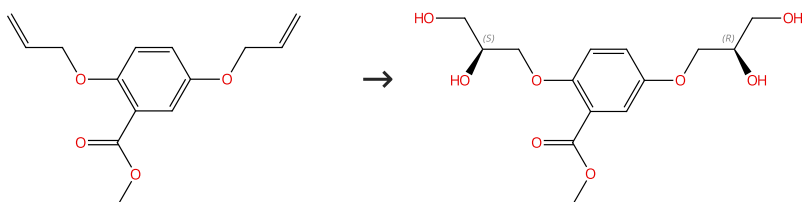
Absolute stereochemistry shown,
Rotation (-)

Suppliers (33)

31-277-CAS-20941319	Steps: 1 Yield: 99%	A Structure-Reactivity Relationship of the Tandem Asymmetric Dihydroxylation on a Biologically Relevant Diene: Influence of Remote Stereocenters on Diastereofacial Selectivity
1.1 Reagents: Potassium carbonate, Potassium ferricyanide Catalysts: Potassium osmate dihydrate, AD-mix-β Solvents: <i>tert</i> -Butanol, Water; rt → 0 °C; 20 min, 0 °C		By: Gill, Daniel M.; et al
1.2 6 h, 0 °C		European Journal of Organic Chemistry (2019), 2019(46), 7568-7577.
1.3 Reagents: Sodium sulfite; 1 h, 0 °C → rt		
Experimental Protocols		

Scheme 12 (1 Reaction)

Steps: 1 Yield: 99%

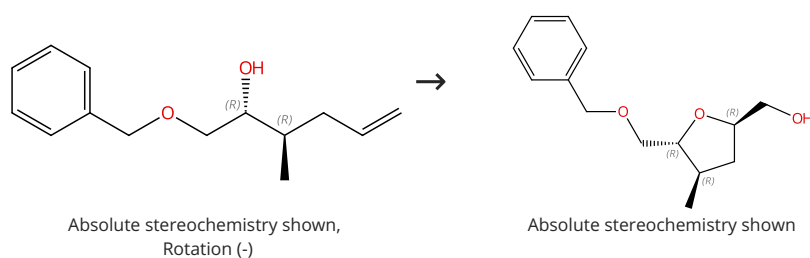
Absolute stereochemistry shown,
Rotation (-)

Suppliers (3)

31-277-CAS-20941315	Steps: 1 Yield: 99%	A Structure-Reactivity Relationship of the Tandem Asymmetric Dihydroxylation on a Biologically Relevant Diene: Influence of Remote Stereocenters on Diastereofacial Selectivity
1.1 Reagents: Potassium carbonate, Potassium ferricyanide Catalysts: Potassium osmate dihydrate, AD-mix- α Solvents: <i>tert</i> -Butanol, Water; rt $\rightarrow$ 0 °C; 20 min, 0 °C		By: Gill, Daniel M.; et al
1.2 6 h, 0 °C		European Journal of Organic Chemistry (2019), 2019(46), 7568-7577.
1.3 Reagents: Sodium sulfite; 1 h, 0 °C $\rightarrow$ rt		
Experimental Protocols		

Scheme 13 (1 Reaction)

Steps: 1 Yield: 99%

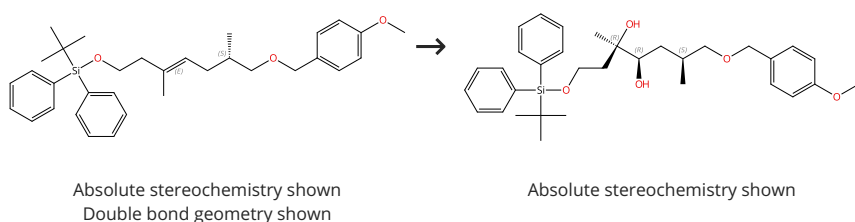


Supplier (1)

31-014-CAS-394464	Steps: 1 Yield: 99%	Gram scale synthesis of the C(1)-C(9) fragment of amphidinolide C
1.1 Reagents: <i>tert</i> -Butyl hydroperoxide, Oxygen Catalysts: Bis[5,5-dimethyl-1-(4-methyl-1-piperazinyl)-1,2,4-hexanetrionato- $\kappa O^2, \kappa O^4$]cobalt Solvents: Isopropanol, 2,2,4-Trimethylpentane; 16 h, 55 °C; 55 °C $\rightarrow$ rt		By: Morra, Nicholas A.; et al
1.2 Reagents: Methyl iodide; 24 h, rt		Tetrahedron (2013), 69(40), 8632-8644.
Experimental Protocols		

Scheme 14 (1 Reaction)

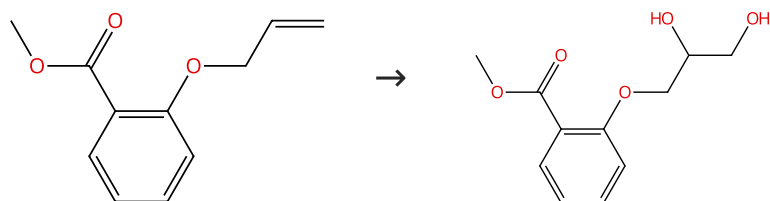
Steps: 1 Yield: 99%



31-277-CAS-255386	Steps: 1 Yield: 99%	Synthesis of amphidinolide Y precursors
1.1 Reagents: Methanesulfonamide Catalysts: AD-mix- β Solvents: <i>tert</i> -Butanol, Water; 48 h, 0 °C		By: Mola, Laura; et al
		Tetrahedron Letters (2014), 55(4), 900-902.

Scheme 15 (1 Reaction)

Steps: 1 Yield: 99%



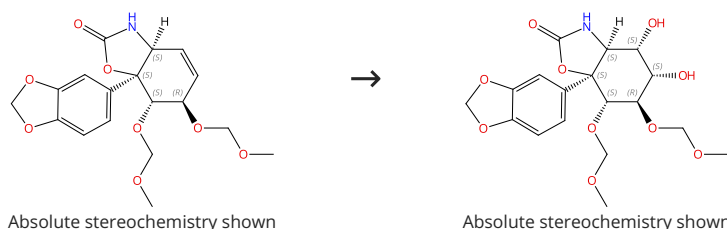
Suppliers (27)

Suppliers (5)

31-277-CAS-20941320	Steps: 1 Yield: 99%	A Structure-Reactivity Relationship of the Tandem Asymmetric Dihydroxylation on a Biologically Relevant Diene: Influence of Remote Stereocenters on Diastereofacial Selectivity
1.1 Reagents: Potassium carbonate, Potassium ferricyanide Catalysts: Potassium osmate dihydrate, AD-mix- α Solvents: <i>tert</i> -Butanol, Water; rt $\rightarrow$ 0 °C; 20 min, 0 °C		By: Gill, Daniel M.; et al
1.2 6 h, 0 °C		European Journal of Organic Chemistry (2019), 2019(46), 7568-7577.
1.3 Reagents: Sodium sulfite; 1 h, 0 °C $\rightarrow$ rt		
Experimental Protocols		

Scheme 16 (1 Reaction)

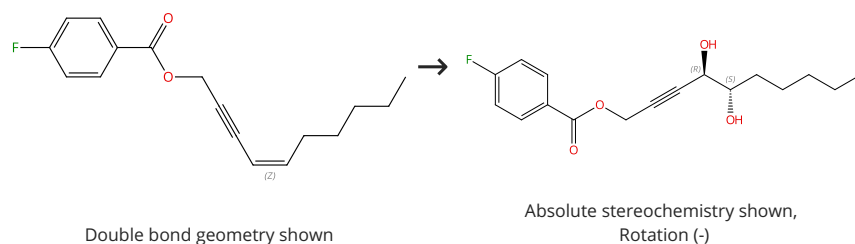
Steps: 1 Yield: 99%



31-277-CAS-6468096	Steps: 1 Yield: 99%	A highly stereocontrolled asymmetric total synthesis of epimer of (+)-7-deoxypancratistatin
1.1 Reagents: N-methylmorpholine N-oxide, Osmium tetroxide Solvents: Acetonitrile, Water; 3 d, rt		By: Chavan, Subhash P.; et al
		Tetrahedron Letters (2013), 54(41), 5562-5566.

Scheme 17 (1 Reaction)

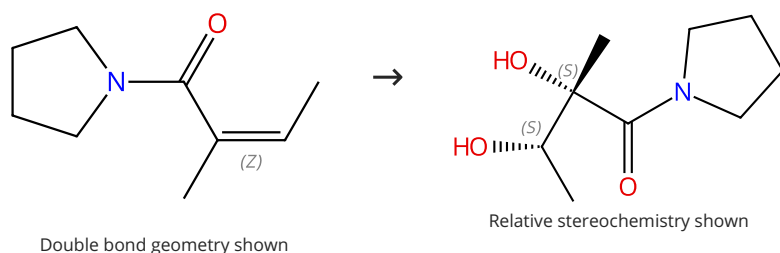
Steps: 1 Yield: 99%



31-277-CAS-20406803	Steps: 1 Yield: 99%	Second-Generation Synthesis of the Northern Fragment of Mandelalide A: Role of π-Stacking on Sharpless Dihydroxylation of <i>cis</i>-Enynes
1.1 Reagents: Potassium carbonate, Methanesulfonamide, Water, Potassium ferricyanide Catalysts: Potassium osmate dihydrate, 1,4-Bis(9- <i>O</i> -dihydroquinidiny)phthalazine Solvents: <i>tert</i> -Butanol; 5 h, rt		By: Ghosh, Ankan; et al
1.2 Reagents: Sodium sulfite; 5 min, rt		Journal of Organic Chemistry (2019), 84(14), 9196-9214.

Scheme 18 (1 Reaction)

Steps: 1 Yield: 99%

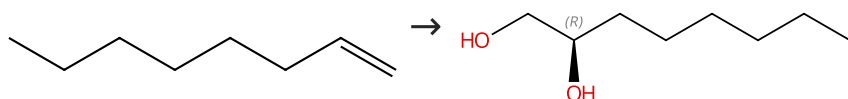


Suppliers (2)

31-277-CAS-10204237	Steps: 1 Yield: 99%	Asymmetric Dihydroxylation of Esters and Amides of Methacrylic, Tiglic, and Angelic Acid: No Exception to the Sharpless Mnemonic!
1.1 Reagents: Citric acid, N-methylmorpholine N-oxide Catalysts: Potassium osmate dihydrate Solvents: <i>tert</i> -Butanol, Water; rt		By: Weber, Fabian; et al
1.2 Reagents: Sodium sulfite Solvents: Water; rt		European Journal of Organic Chemistry (2015), 2015(11), 2428-2449.
Experimental Protocols		

Scheme 19 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (93)

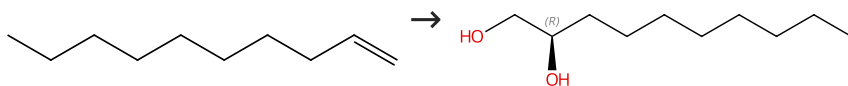
Absolute stereochemistry shown,
Rotation (+)

Suppliers (13)

31-277-CAS-6068690	Steps: 1 Yield: 99%	Total synthesis of mniopetals A, B, C and D
1.1 Reagents: AD-mix-β Solvents: <i>tert</i> -Butanol, Water		By: Weihrather, Joachim; et al Synlett (2013), 24(11), 1410-1414.

Scheme 20 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (76)

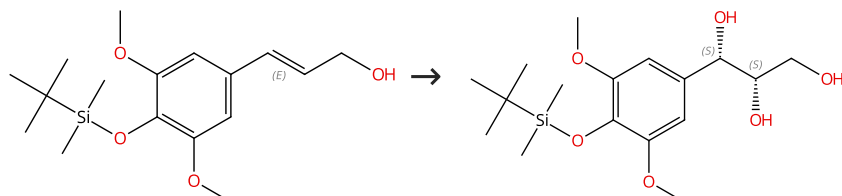
Absolute stereochemistry shown,
Rotation (+)

Suppliers (18)

31-277-CAS-15302630	Steps: 1 Yield: 99%	Total synthesis of mniopetals A, B, C and D
1.1 Reagents: AD-mix-β Solvents: <i>tert</i> -Butanol, Water		By: Weihrather, Joachim; et al Synlett (2013), 24(11), 1410-1414.

Scheme 21 (1 Reaction)

Steps: 1 Yield: 98%



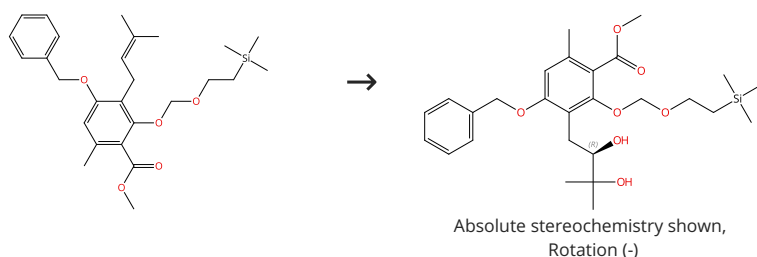
Double bond geometry shown

Absolute stereochemistry shown

31-614-CAS-41269276	Steps: 1 Yield: 98%	Modular Enantioselective Total Syntheses of the erythro-7,9-Dihydroxy- and 9-Hydroxy-7-Keto-8,4'-Oxyneolignans
1.1 Reagents: Methanesulfonamide, AD-mix- α Solvents: <i>tert</i> -Butanol, Water; rt $\rightarrow$ 0 °C		By: Benda, Meghan C.; et al
1.2 0 °C $\rightarrow$ rt; overnight, rt		Journal of Organic Chemistry (2024), 89(14), 9910-9922.
1.3 Reagents: Sodium sulfite; 30 min, rt		
Experimental Protocols		

Scheme 22 (1 Reaction)

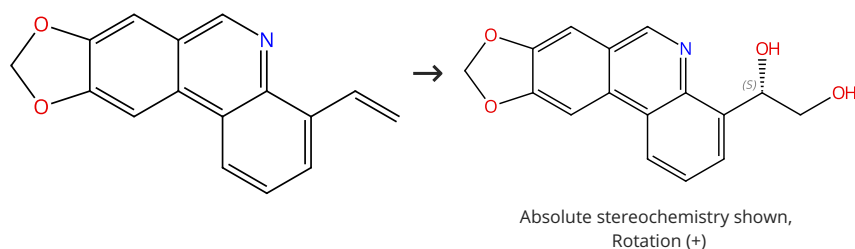
Steps: 1 Yield: 98%



31-614-CAS-32029991	Steps: 1 Yield: 98%	The Asymmetric Total Synthesis of (-)-Eurothiocin A and Its Enantiomer
1.1 Reagents: Methanesulfonamide Solvents: <i>tert</i> -Butanol, Water; 0 °C; 5 h, rt		By: Zhang, Guangyan; et al
1.2 Reagents: Sodium sulfite Solvents: Water; rt		Journal of Natural Products (2022), 85(4), 997-1005.
Experimental Protocols		

Scheme 23 (1 Reaction)

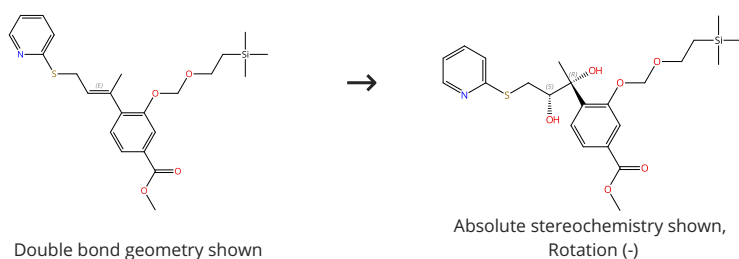
Steps: 1 Yield: 98%



31-277-CAS-20761018	Steps: 1 Yield: 98%	First asymmetric enantioselective total synthesis of phenanthridine alkaloid, (S)-(+)-asiaticumine and its enantiomer
1.1 Catalysts: Methanesulfonamide, AD-mix- α Solvents: <i>tert</i> -Butanol, Water; 0 °C; 24 h, rt		By: Nishiyama, Takashi; et al
1.2 Reagents: Sodium sulfite; 30 min, rt		Tetrahedron Letters (2019), 60(48), 151278.
Experimental Protocols		

Scheme 24 (1 Reaction)

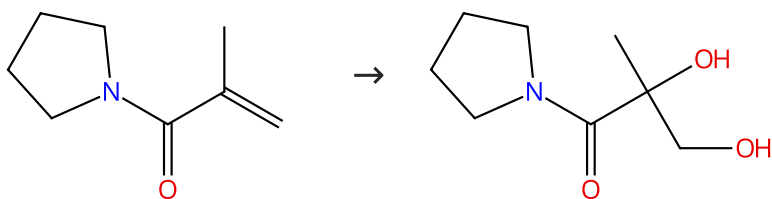
Steps: 1 Yield: 98%



31-277-CAS-24351467	Steps: 1 Yield: 98%	Practical synthesis of aromatic bisabolanes: Synthesis of 1,3,5-bisabolatrien-7-ol, peniciaculin A and B, and hydroxysydonic acid
1.1 Reagents: Methanesulfonamide, AD-mix- β Solvents: <i>tert</i> -Butanol, Water; 22 h, 4 °C		By: Yajima, Arata; et al
1.2 Reagents: Sodium thiosulfate Solvents: Water		Tetrahedron (2021), 92, 132253.
Experimental Protocols		

Scheme 25 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (5)

31-277-CAS-9389897

Steps: 1 Yield: 98%

Asymmetric Dihydroxylation of Esters and Amides of Methacrylic, Tiglic, and Angelic Acid: No Exception to the Sharpless Mnemonic!

By: Weber, Fabian; et al

European Journal of Organic Chemistry (2015), 2015(11), 2428-2449.

1.1 Reagents: Citric acid, N-methylmorpholine N-oxide

Catalysts: Potassium osmate dihydrate

Solvents: *tert*-Butanol, Water; rt

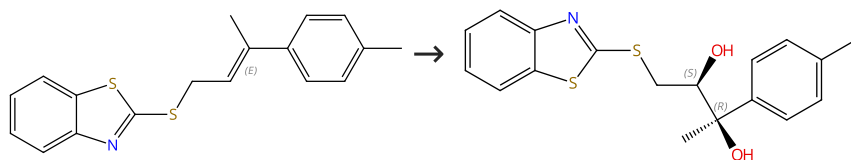
1.2 Reagents: Sodium sulfite

Solvents: Water; rt

Experimental Protocols

Scheme 26 (1 Reaction)

Steps: 1 Yield: 98%



Double bond geometry shown

Absolute stereochemistry shown,
Rotation (-)

31-277-CAS-24351106

Steps: 1 Yield: 98%

Practical synthesis of aromatic bisabolanes: Synthesis of 1,3,5-bisabolatrien-7-ol, peniciculin A and B, and hydroxysydonic acid

By: Yajima, Arata; et al

Tetrahedron (2021), 92, 132253.

1.1 Reagents: Methanesulfonamide, AD-mix-β

Solvents: *tert*-Butanol, Water; 0 °C; 38 h, 4 °C

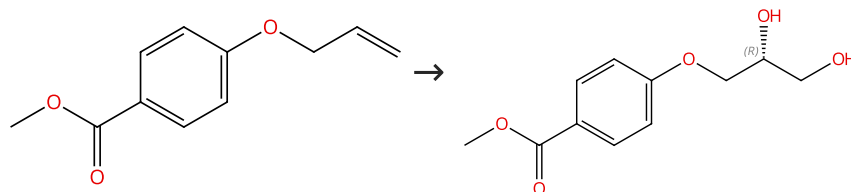
1.2 Reagents: Sodium sulfite

Solvents: Water

Experimental Protocols

Scheme 27 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (33)

Absolute stereochemistry shown,
Rotation (+)

31-277-CAS-20941318

Steps: 1 Yield: 97%

A Structure-Reactivity Relationship of the Tandem Asymmetric Dihydroxylation on a Biologically Relevant Diene: Influence of Remote Stereocenters on Diastereofacial Selectivity

By: Gill, Daniel M.; et al

European Journal of Organic Chemistry (2019), 2019(46), 7568-7577.

1.1 Reagents: Potassium carbonate, Potassium ferricyanide

Catalysts: Potassium osmate dihydrate, AD-mix-α

Solvents: *tert*-Butanol, Water; rt → 0 °C; 20 min, 0 °C

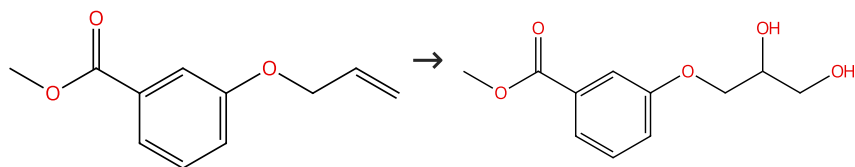
1.2 6 h, 0 °C

1.3 Reagents: Sodium sulfite; 1 h, 0 °C → rt

Experimental Protocols

Scheme 28 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (28)

Suppliers (9)

31-277-CAS-20941332

Steps: 1 Yield: 97%

- 1.1 **Reagents:** N-methylmorpholine N-oxide
Catalysts: Potassium osmate dihydrate
Solvents: Acetone, Water; 10 min, rt
- 1.2 rt; 8 h, 40 °C
- 1.3 **Reagents:** Sodium sulfite; 1 h, 40 °C → rt

Experimental Protocols

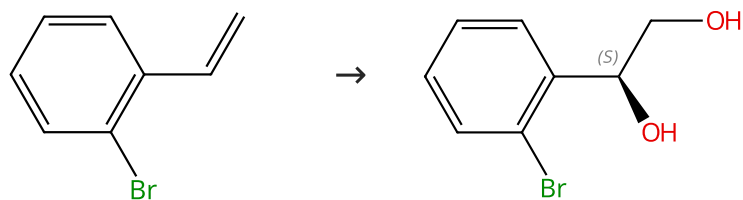
A Structure-Reactivity Relationship of the Tandem Asymmetric Dihydroxylation on a Biologically Relevant Diene: Influence of Remote Stereocenters on Diastereofacial Selectivity

By: Gill, Daniel M.; et al

European Journal of Organic Chemistry (2019), 2019(46), 7568-7577.

Scheme 29 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (76)

Absolute stereochemistry shown,
Rotation (+)

Suppliers (6)

31-277-CAS-12541974

Steps: 1 Yield: 97%

- 1.1 **Reagents:** AD-mix- α
Solvents: *tert*-Butanol, Water; 21.5 h, 0 °C
- 1.2 **Reagents:** Sodium sulfite; 0 °C → rt; 40 min, rt

Experimental Protocols

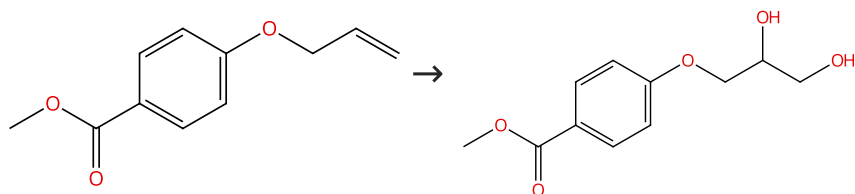
Sharpless asymmetric dihydroxylation as the key step in the enantioselective synthesis of spirocyclic σ_1 receptor ligands

By: Holl, Katharina; et al

Tetrahedron: Asymmetry (2014), 25(3), 268-277.

Scheme 30 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (33)

Supplier (1)

31-277-CAS-20941333

Steps: 1 Yield: 97%

- 1.1 **Reagents:** N-methylmorpholine N-oxide
Catalysts: Potassium osmate dihydrate
Solvents: Acetone, Water; 10 min, rt
- 1.2 rt; 8 h, 40 °C
- 1.3 **Reagents:** Sodium sulfite; 1 h, 40 °C → rt

Experimental Protocols

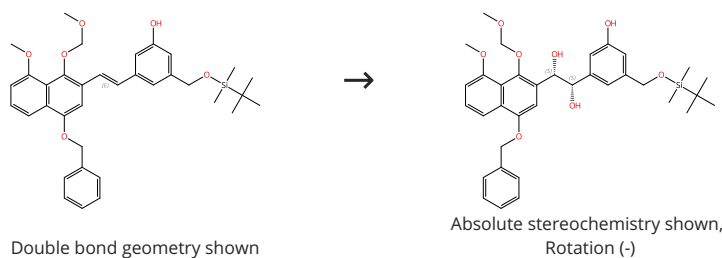
A Structure-Reactivity Relationship of the Tandem Asymmetric Dihydroxylation on a Biologically Relevant Diene: Influence of Remote Stereocenters on Diastereofacial Selectivity

By: Gill, Daniel M.; et al

European Journal of Organic Chemistry (2019), 2019(46), 7568-7577.

Scheme 31 (1 Reaction)

Steps: 1 Yield: 97%



31-277-CAS-23207176

Steps: 1 Yield: 97%

Asymmetric Total Synthesis of PD-116740

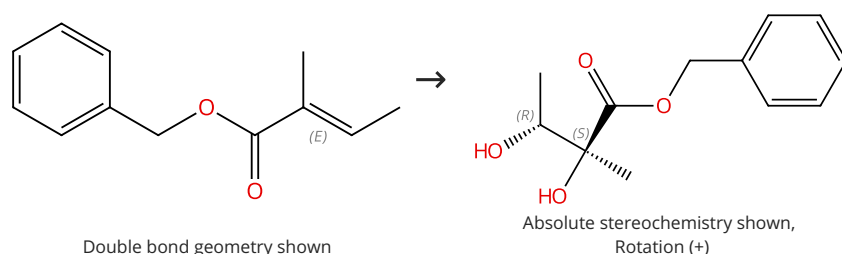
- 1.1 **Reagents:** Osmium tetroxide, (DHQ)₂PHAL
Solvents: Dichloromethane, Toluene; -78 °C; -78 °C → -24 °C; 4 h, -24 °C; -24 °C → -78 °C
- 1.2 **Reagents:** Hydrogen sulfide; 10 min, -78 °C; 30 min, -78 °C

By: Zheng, Chaoying; et al
 Organic Letters (2021), 23(2), 469-473.

Experimental Protocols

Scheme 32 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (63)

Suppliers (10)

31-614-CAS-33736012

Steps: 1 Yield: 97%

Synthesis and Bioactivity of Ophiofuranones A and B

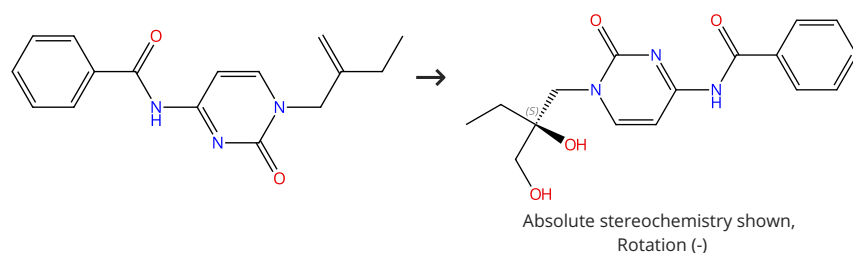
- 1.1 **Reagents:** Methanesulfonamide
Catalysts: AD-mix-β
Solvents: *tert*-Butanol, Water; 5 d, 0 °C
- 1.2 **Reagents:** Sodium sulfite; 0 °C → rt; 1 h, rt

By: Gillsch, Franziska; et al
 Journal of Organic Chemistry (2022), 87(9), 6520-6523.

Experimental Protocols

Scheme 33 (1 Reaction)

Steps: 1 Yield: 97%



31-277-CAS-19450365

Steps: 1 Yield: 97%

Synthesis of Chiral Acyclic Nucleosides by Sharpless Asymmetric Dihydroxylation: Access to Cidofovir and Buciclovir

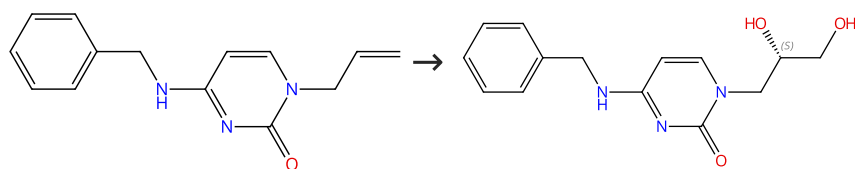
- 1.1 **Reagents:** Potassium carbonate, Methanesulfonamide, Potassium ferricyanide
Catalysts: Potassium osmate dihydrate, 1,4-Bis(9-*O*-dihydroquinidiny)phthalazine
Solvents: *tert*-Butanol, Water; 30 min, rt
- 1.2 12 h, 0 °C
- 1.3 **Reagents:** Sodium thiosulfate; 0 °C; 2 h, rt

By: Qin, Tao; et al
 Journal of Organic Chemistry (2018), 83(24), 15512-15523.

Experimental Protocols

Scheme 34 (1 Reaction)

Steps: 1 Yield: 96%

Absolute stereochemistry shown,
Rotation (-)

31-277-CAS-19450363

Steps: 1 Yield: 96%

Synthesis of Chiral Acyclic Nucleosides by Sharpless Asymmetric Dihydroxylation: Access to Cidofovir and Buciclovir

By: Qin, Tao; et al

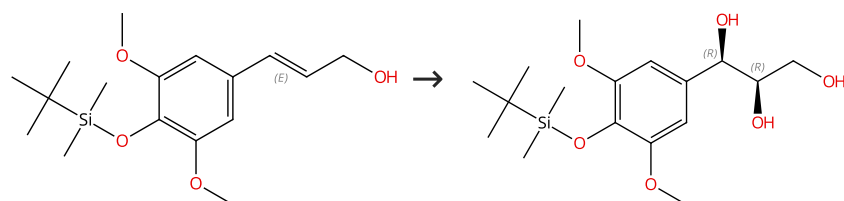
Journal of Organic Chemistry (2018), 83(24), 15512-15523.

- 1.1 **Reagents:** Potassium carbonate, Methanesulfonamide, Potassium ferricyanide
Catalysts: Potassium osmate dihydrate, 1,4-Bis(9-*O*-dihydroquinidinyl)phthalazine
Solvents: *tert*-Butanol, Water; 30 min, rt
- 1.2 12 h, 0 °C
- 1.3 **Reagents:** Sodium thiosulfate; 0 °C; 2 h, rt

Experimental Protocols

Scheme 35 (1 Reaction)

Steps: 1 Yield: 96%



Double bond geometry shown

Absolute stereochemistry shown

31-614-CAS-41269274

Steps: 1 Yield: 96%

Modular Enantioselective Total Syntheses of the erythro-7,9-Dihydroxy- and 9-Hydroxy-7-Keto-8,4'-Oxyneolignans

By: Benda, Meghan C.; et al

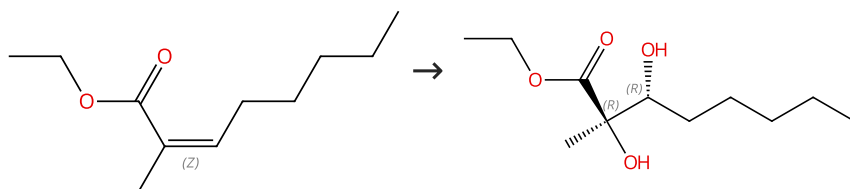
Journal of Organic Chemistry (2024), 89(14), 9910-9922.

- 1.1 **Reagents:** Methanesulfonamide, AD-mix-β
Solvents: *tert*-Butanol, Water; rt → 0 °C
- 1.2 0 °C → rt; overnight, rt
- 1.3 **Reagents:** Sodium sulfite; 30 min, rt

Experimental Protocols

Scheme 36 (1 Reaction)

Steps: 1 Yield: 96%



Double bond geometry shown

Absolute stereochemistry shown,
Rotation (+)

Suppliers (2)

31-277-CAS-3959702

Steps: 1 Yield: 96%

Establishment of Relative and Absolute Configurations of Phaeosphaeride A: Total Synthesis of ent- Phaeosphaeride A

By: Kobayashi, Kenichi; et al

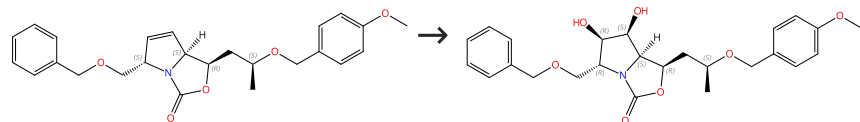
Journal of Organic Chemistry (2015), 80(2), 1243-1248.

- 1.1 **Reagents:** Methanesulfonamide
Catalysts: AD-mix-β
Solvents: *tert*-Butanol, Water; 17 h, 0 °C
- 1.2 **Reagents:** Sodium sulfite
Solvents: Water; 0 °C; 0 °C → rt; 1 h, rt

Experimental Protocols

Scheme 37 (1 Reaction)

Steps: 1 Yield: 96%

Absolute stereochemistry shown,
Rotation (-)Absolute stereochemistry shown,
Rotation (+)

31-277-CAS-14850572

Steps: 1 Yield: 96%

Total Synthesis of Hyacinthacines B₃, B₄, and B₅ and
Purported Hyacinthacine B₇, 7-epi-Hyacinthacine B₇, and 7a-
epi-Hyacinthacine B₃ from a Common Precursor

By: Savasapun, Kongdech; et al

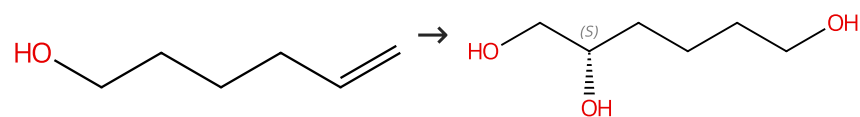
Journal of Organic Chemistry (2014), 79(10), 4569-4581.

1.1 Reagents: N-methylmorpholine N-oxide
Catalysts: Dipotassium osmate
Solvents: Acetone, Water; 18 h, 35 °C

Experimental Protocols

Scheme 38 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (88)

Absolute stereochemistry shown,
Rotation (-)

Suppliers (4)

31-277-CAS-273039

Steps: 1 Yield: 96%

First stereoselective total synthesis of triumfettamide

By: Devi, Thongam Joymati; et al

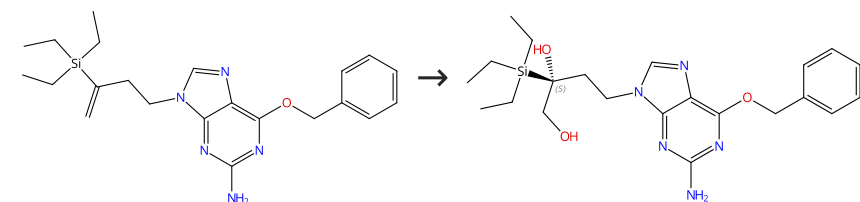
Tetrahedron (2013), 69(45), 9457-9462.

1.1 Reagents: AD-mix-β
Solvents: *tert*-Butanol, Water; 16 h, 0 °C
1.2 Reagents: Sodium pyrosulfite; 0 °C; 0 °C → rt; 1 h, rt

Experimental Protocols

Scheme 39 (1 Reaction)

Steps: 1 Yield: 96%

Absolute stereochemistry shown,
Rotation (+)

31-277-CAS-19450386

Steps: 1 Yield: 96%

Synthesis of Chiral Acyclic Nucleosides by Sharpless
Asymmetric Dihydroxylation: Access to Cidofovir and
Buciclovir

By: Qin, Tao; et al

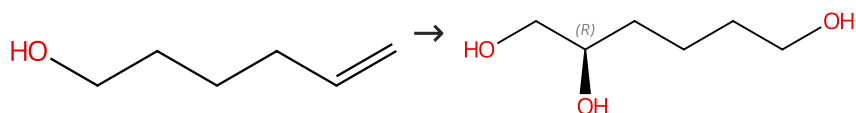
Journal of Organic Chemistry (2018), 83(24), 15512-15523.

1.1 Reagents: Potassium carbonate, Methanesulfonamide,
Potassium ferricyanide
Catalysts: Potassium osmate dihydrate, 1,4-Bis[(9S)-10,11-
dihydro-6'-methoxycinchonan-9-yl]-9,10-anthracenedione
Solvents: *tert*-Butanol, Water; 30 min, rt
1.2 12 h, 0 °C
1.3 Reagents: Sodium thiosulfate; 0 °C; 2 h, rt

Experimental Protocols

Scheme 40 (1 Reaction)

Steps: 1 Yield: 96%

Absolute stereochemistry shown,
Rotation (+)

Suppliers (88)

Suppliers (4)

31-277-CAS-11040808

Steps: 1 Yield: 96%

First stereoselective total synthesis of triumfettamide

1.1 Reagents: AD-mix-α

Solvents: *tert*-Butanol, Water; 16 h, 0 °C

1.2 Reagents: Sodium pyrosulfite; 0 °C; 0 °C → rt; 1 h, rt

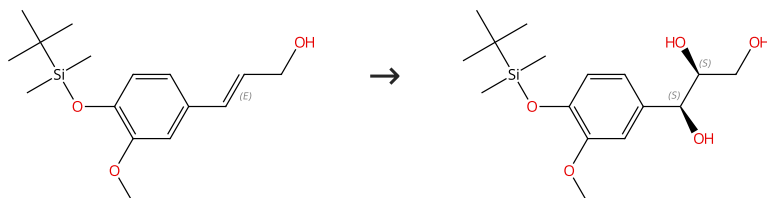
Experimental Protocols

By: Devi, Thongam Joymati; et al

Tetrahedron (2013), 69(45), 9457-9462.

Scheme 41 (1 Reaction)

Steps: 1 Yield: 96%



Double bond geometry shown

Absolute stereochemistry shown

Supplier (1)

31-614-CAS-41269278

Steps: 1 Yield: 96%

Modular Enantioselective Total Syntheses of the erythro-7,9-Dihydroxy- and 9-Hydroxy-7-Keto-8,4'-Oxyneolignans

1.1 Reagents: Methanesulfonamide, AD-mix-α

Solvents: *tert*-Butanol, Water; rt → 0 °C

1.2 0 °C → rt; overnight, rt

1.3 Reagents: Sodium sulfite; 30 min, rt

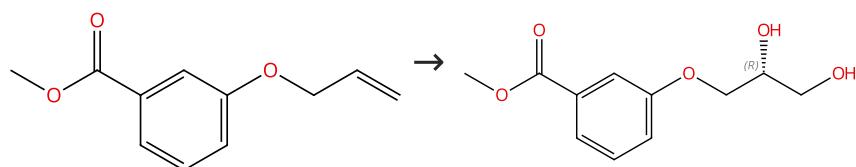
Experimental Protocols

By: Benda, Meghan C.; et al

Journal of Organic Chemistry (2024), 89(14), 9910-9922.

Scheme 42 (1 Reaction)

Steps: 1 Yield: 96%



Absolute stereochemistry shown

Suppliers (28)

31-277-CAS-20941317

Steps: 1 Yield: 96%

A Structure-Reactivity Relationship of the Tandem Asymmetric Dihydroxylation on a Biologically Relevant Diene: Influence of Remote Stereocenters on Diastereofacial Selectivity

1.1 Reagents: Potassium carbonate, Potassium ferricyanide

Catalysts: Potassium osmate dihydrate, AD-mix-α

Solvents: *tert*-Butanol, Water; rt → 0 °C; 20 min, 0 °C

1.2 6 h, 0 °C

1.3 Reagents: Sodium sulfite; 1 h, 0 °C → rt

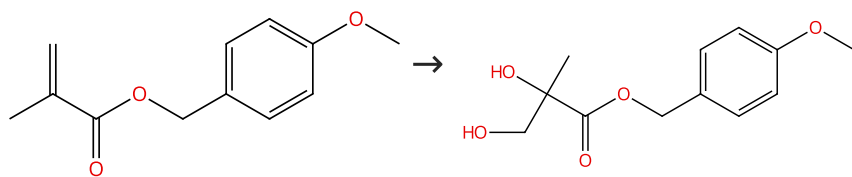
Experimental Protocols

By: Gill, Daniel M.; et al

European Journal of Organic Chemistry (2019), 2019(46), 7568-7577.

Scheme 43 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (2)

31-277-CAS-4940828

Steps: 1 Yield: 96%

Asymmetric Dihydroxylation of Esters and Amides of Methacrylic, Tiglic, and Angelic Acid: No Exception to the Sharpless Mnemonic!

By: Weber, Fabian; et al

European Journal of Organic Chemistry (2015), 2015(11), 2428-2449.

1.1 Reagents: Citric acid, N-methylmorpholine N-oxide

Catalysts: Potassium osmate dihydrate

Solvents: *tert*-Butanol, Water; rt

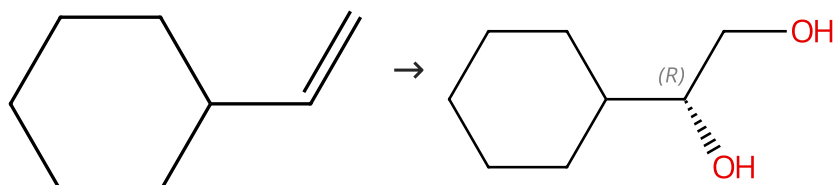
1.2 Reagents: Sodium sulfite

Solvents: Water; rt

Experimental Protocols

Scheme 44 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (54)

Absolute stereochemistry shown,
Rotation (-)

Suppliers (4)

31-277-CAS-21822510

Steps: 1 Yield: 96%

Absolute stereochemical determination of 1,2-diols via complexation with dinaphthyl borinic acid

By: Torabi Kohlbouni, Saeedeh; et al

Chirality (2020), 32(6), 817-823.

1.1 Reagents: Methanesulfonamide, AD-mix-β

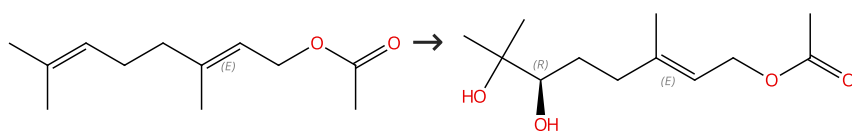
Solvents: *tert*-Butanol, Water; 30 min, 4 °C

1.2 24 h, 4 °C

1.3 Reagents: Sodium sulfite; 45 min, rt

Scheme 45 (1 Reaction)

Steps: 1 Yield: 96%



Double bond geometry shown

Suppliers (87)

Absolute stereochemistry shown,
Rotation (+)

Double bond geometry shown

Suppliers (2)

31-277-CAS-19243525

Steps: 1 Yield: 96%

Asymmetric synthesis and structure-activity studies of the fungal metabolites colletorin A, colletochlorin A and their halogenates analogues

By: Marsico, Giulia; et al

Tetrahedron (2018), 74(28), 3912-3923.

1.1 Reagents: Methanesulfonamide, AD-mix-β

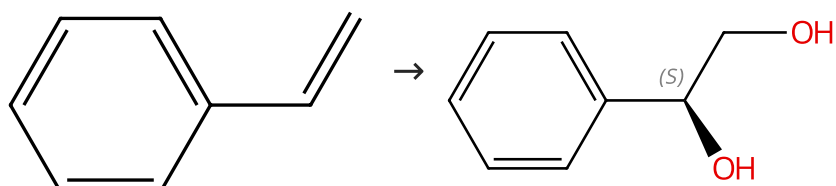
Solvents: *tert*-Butanol, Water; 15 min, rt; rt → 0 °C

1.2 24 h, 0 °C

1.3 Reagents: Sodium pyrosulfite

Scheme 46 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (115)

Absolute stereochemistry shown,
Rotation (+)

Suppliers (79)

31-277-CAS-13126816

Steps: 1 Yield: 96%

Application of Sharpless asymmetric dihydroxylation to
synthesis of chiral β -amino alcohols

No Data Available

Experimental Protocols

By: Liu, Wen-Min; et al

Youji Huaxue (2006), 26(3), 341-345.

Scheme 47 (1 Reaction)

Steps: 1 Yield: 95%

Absolute stereochemistry shown,
Rotation (+)
Double bond geometry shownAbsolute stereochemistry shown,
Rotation (+)

31-277-CAS-14774972

Steps: 1 Yield: 95%

A stereoselective approach to indolizidine and pyrrolizidine
alkaloids: total synthesis of (-)-lentiginosine, (-)-epi-lentigi
nosine and (-)-dihydroxypyrrolizidine

1.1 **Reagents:** Potassium carbonate, Potassium ferricyanide
Catalysts: Methanesulfonamide, Osmium tetroxide, 1,4-Bis(9-*O*-dihydroquinidiny)phthalazine
Solvents: *tert*-Butanol, Toluene, Water; 5 min, 0 °C

1.2 24 h, 0 °C

1.3 **Reagents:** Sodium sulfite; 15 min, 0 °C

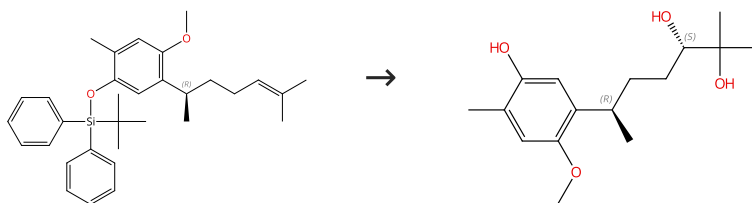
Experimental Protocols

By: Kauloorkar, Shruti Vandana; et al

Organic & Biomolecular Chemistry (2014), 12(25), 4454-4460.

Scheme 48 (1 Reaction)

Steps: 1 Yield: 95%



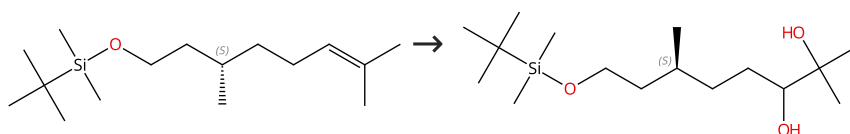
Absolute stereochemistry shown

Absolute stereochemistry shown,
Rotation (-)

31-049-CAS-9729608	Steps: 1 Yield: 95%	Total synthesis of heliespirone B
1.1 Reagents: Potassium carbonate, Potassium ferricyanide Catalysts: Dipotassium osmate, 1,4-Bis(9- <i>O</i> -dihydroquinidinyI) phthalazine Solvents: <i>tert</i> -Butanol, Water; 2 h, rt		By: Yang, Li; et al
1.2 Reagents: Methanesulfonamide; 30 min, rt		Tetrahedron Letters (2015), 56(34), 4931-4933.
1.3 Solvents: <i>tert</i> -Butanol; overnight, 0 °C		
1.4 Reagents: Tetrabutylammonium fluoride Solvents: Water; 5 h, 0 °C		
1.5 Reagents: Water; 0 °C		
Experimental Protocols		

Scheme 49 (1 Reaction)

Steps: 1 Yield: 95%

Absolute stereochemistry shown,
Rotation (-)

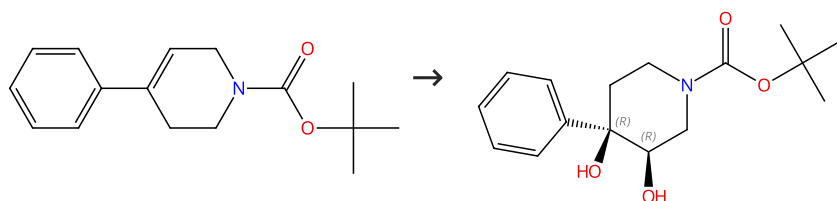
Absolute stereochemistry shown

Suppliers (3)

31-277-CAS-16265424	Steps: 1 Yield: 95%	Effective method to C(10)-C(22) fragment of leiodelide A
1.1 Reagents: N-methylmorpholine N-oxide, Dipotassium osmate Solvents: <i>tert</i> -Butanol, Water; 24 h, 0 °C		By: Ren, Rongguo; et al
1.2 Reagents: Sodium sulfite Solvents: Water		Youji Huaxue (2015), 35(11), 2313-2320.

Scheme 50 (1 Reaction)

Steps: 1 Yield: 95%

Absolute stereochemistry shown,
Rotation (+)

Suppliers (37)

Supplier (1)

31-277-CAS-8689014	Steps: 1 Yield: 95%	Synthesis of Fused Bicyclic Piperi dines: Potential Bioactive Templates for Medicinal Chemistry
1.1 Reagents: AD-mix-β		By: Zhou, Jinglan; et al
		Journal of Organic Chemistry (2015), 80(1), 70-79.